VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE
Near Ramoji Film City, Deshmukhi Village, Pochampally Mandal, Yadadri Bhuvanagiri Dist.
(Approved by AICTE, New Delhi, Affiliated to JNTUH, Hyderabad)
AN AUTONOMOUS INSTITUTION

Eamcet Code VGNT Dist Code : YBG

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI & ML)

YEAR & SEM: IV – II **AY: 2025-26**
SECTION: A

MAJOR PROJECT

BATCH-A13

TITLE OF PROJECT:

CrossTongueGuard: Detecting Cyberbullying in Code-Mixed Conversations.

ROLL NUMBER	STUDENT NAME	PH.NO	EMAIL ID
22891A6606	B. Denesh Kumar	7569888133	deneshkumar248@gmail.com
22891A6622	K. Srinadh	8688338093	srinadhkaramkonda@gmail.com
22891A6639	N. Upender Goud	8712285235	nakkaupendergoud@gmail.com
22891A6643	P. Bhaskar	6302817510	bhaskarabdj123@gmail.com

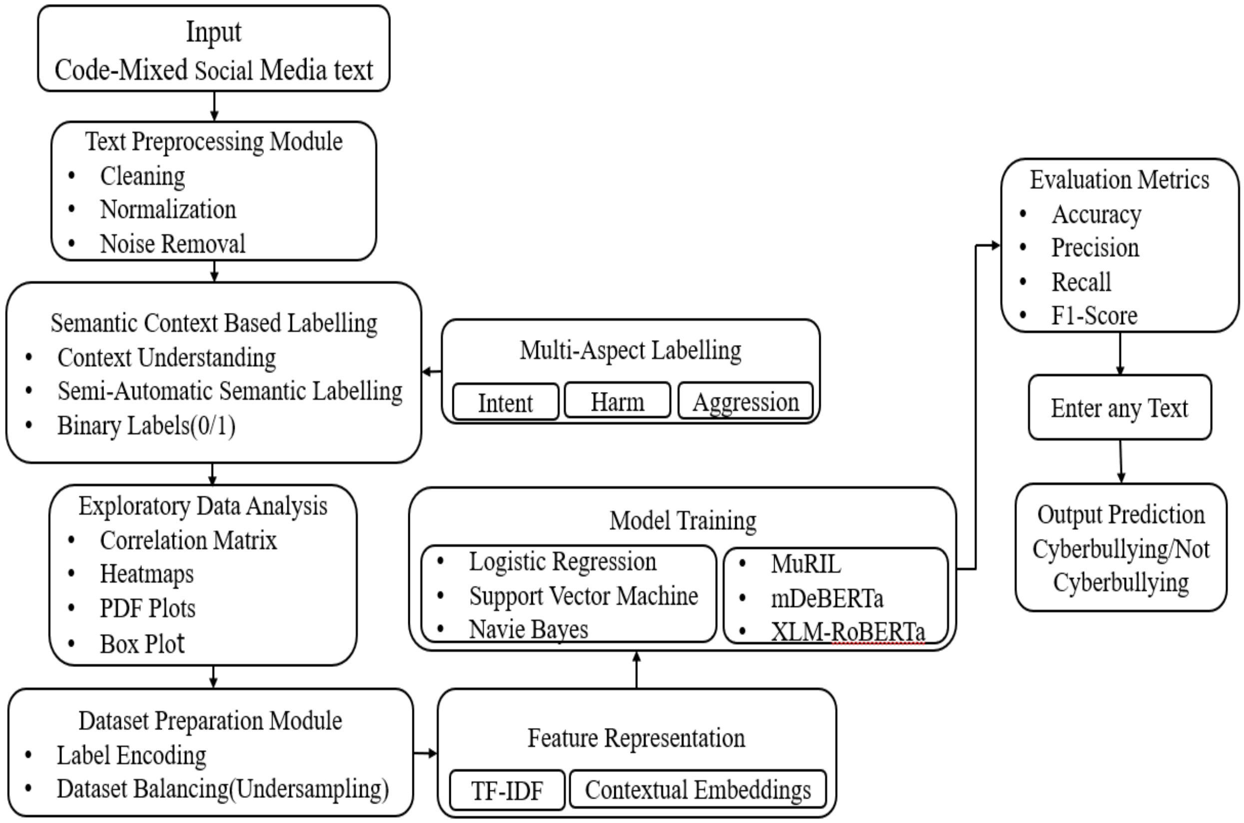
GUIDE NAME: Mr. K. Sudhakar

ABSTRACT

Cyberbullying has emerged as a recent concern in the global online communication platform, causing emotional distress and psychological trauma to individuals. The detection of cyberbullying is particularly challenging in the multilingual code-mixed online social media communication platform, where individuals communicate in multiple languages in a single message and use informal language, transliteration, and regional slang. These language complexities reduce the efficiency of traditional text classification techniques. To address this challenge, this paper proposes CrossTongueGuard, a systematic framework for the detection of cyberbullying in multilingual code-mixed communication platforms. The framework employs systematic preprocessing techniques, semantic context-based annotation, and multi-aspect analysis based on intent, harm, and aggression. The framework integrates traditional machine learning models and multilingual transformer models to efficiently extract both statistical and contextual features. The primary aim of this research is to develop a scalable and language-agnostic cyberbullying detection framework that can handle real-world multilingual social media data efficiently and establish a safe online communication environment.

Keywords—Code-Mixed, Semantic annotation, Multilingual Transformer Models, Contextual Features.

BLOCK DIAGRAM / CIRCUIT DIAGRAM:



GUIDE NAME &SIGNATURE **HOD-CSE(AI&ML)**